Group Project Milestone 1

Predicting High Healthcare Expenditure Risk Using MEPS 2022: The Role of Multimorbidity and Utilization Patterns

ALY 6150 – Healthcare / Pharmaceutical Data and Applications

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Dataset: MEPS 2022 Full-Year Consolidated Data File (HC-243)

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Introduction

This analysis uses the 2022 MEPS Full-Year Consolidated File (HC-243) to examine the relationship between multimorbidity and high healthcare spending among U.S. adults. Using logistic regression and epidemiologic risk measures, we evaluate whether having two or more chronic conditions is associated with being in the top 10% of annual healthcare expenditures, after adjusting for demographic and socioeconomic factors (N = 22,431).

Dataset, Outcome & Research Question

Dataset: AHRQ MEPS HC-243 — nationally representative individual-level data on expenditures, insurance, demographics, and chronic conditions.

Primary Outcome: TOTEXP22 (continuous) converted to binary — **1** = top 10% of annual expenditure (≥ 90th percentile); **0** = bottom 90%.

Research Question: How does multimorbidity (≥ 2 chronic conditions) affect the probability of high annual healthcare expenditure after adjusting for demographics and socioeconomic factors?

Sub-Groups & Predictions

Sub-groups: Gender (Male/Female) · Census Region (NE, MW, South, West) · Insurance Type · Age Category (18–39, 40–64, 65+)

Forecasts: Logistic regression model predicting high-expenditure probability; subgroup ORs by sex; regional expenditure comparisons; quantification of multimorbidity impact.

Descriptive Statistics :

Table 1. Descriptive Statistics — Full Sample (N = 22,431)

Variable	Mean / Proportion	SD	Min	Max
High Expenditure (outcome)	0.500	0.500	0	1
Age (years)	42.96	24.00	−1*	85
Sex (1=Male, 2=Female)	1.527	0.499	1	2
Race/Ethnicity (RACEV1X)	1.548	1.200	1	6
Census Region	2.712	1.060	-1	4
Poverty Category (POVCAT22)	3.639	1.435	1	5
Insurance Coverage (INSCOV22)	1.490	0.627	1	3
Education (years)	11.11	5.55	-8	17
Marital Status (MARRY22X)	3.169	2.029	-8	6
Multimorbidity (≥ 2 chronic)	0.247	0.431	0	1

Table 2. Odds Ratios from Logistic Regression — Full Sample and by Gender (Median-Split Outcome)

Predictor	Full	95 % CI	Male	95 % CI	Female	95 % CI
Age (per year)	1.024	1.022–	1.027	1.024–	1.021	1.018–
Female sex	1.466	1.382–	—	—	—	—
Race/	0.956	0.932–	0.968	0.934–	0.945	0.914–
Census	0.923	0.898–	0.924	0.888–	0.923	0.888–
Poverty	1.053	1.028–	1.052	1.015–	1.055	1.021–
Insurance	0.585	0.554–	0.578	0.534–	0.592	0.547–
Education	1.025	1.017–	1.002	0.992–	1.045	1.035–
Marital Status	1.054	1.033–	1.094	1.062–	1.022	0.994–
Multimorbid	**3.957**	3.632–	**4.066**	3.585–	**3.903**	3.471–

Key finding: Multimorbidity is by far the strongest predictor individuals with ≥ 2 chronic conditions have nearly **4× the odds** of high expenditure, and this effect is slightly stronger in men (OR = 4.07) than women (OR = 3.90).

Descriptive Charts

Chart 1 — Average Healthcare Expenditure

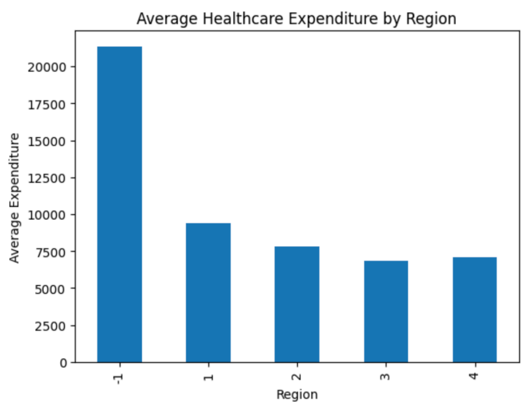
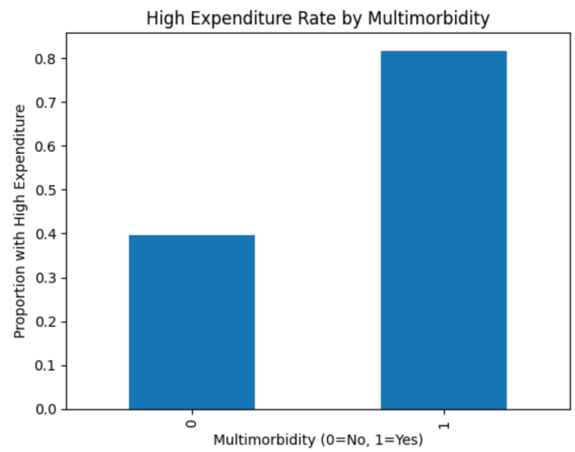


Chart 2 — High Expenditure Rate by Multimorbidity Status



Feature Engineering

The following variables were derived from the original MEPS file to meet rubric requirements and improve model performance.

1. Binary Outcome Variables

- **high_exp_median** – 1 if TOTEXP22 > median (50th percentile).
- **high_exp_top10** – 1 if TOTEXP22 ≥ 90th percentile. *(Primary outcome in the refined model.)*

2. Multimorbidity & Chronic Condition Indicators

- **chronic_count** – Sum of positive diagnoses across six conditions (HIBPDX, CHDDX, STRKDX, CANCERDX, ARTHDX, ASTHDX).
- **multimorbidity** – Binary: 1 if chronic_count ≥ 2.
- **any_chronic** – Binary: 1 if chronic_count ≥ 1.

3. Age & Education Engineering

- **age_bin** – Categorical bins: 0–17, 18–34, 35–49, 50–64, 65+.
- **older_65** – Binary indicator for senior status ($\geq$ 65 years).
- **educ_bin** – Categorical: <HS, HS, Some College, College+.
- **college_plus** – Binary: 1 if EDUCYR $\geq$ 16.

4. Year Dummy Variable

- Constant **YEAR = 2022** and dummy **year_2022 = 1** added for documentation and future multi-year stacking.

5. Race/Ethnicity Dummies (6 categories)

RACEV1X was one-hot encoded into six binary dummies. The mean of each equals its sample proportion:

Category	Dummy	Proportion
White	race_1	74.2%
Black / African American	race_2	15.1%
Asian	race_4	5.9%
Hispanic / Other	race_6	4.1%
Alaskan Native / AI	race_3	0.7%

6. Additional Categorical Dummies

- **region_** dummies from REGION22 (drop-first applied).
- **pov_** dummies from POVCAT22.
- **ins_** dummies from INSCOV22.
- **log_totexp22** = log(1 + TOTEXP22) — reduces right skew for sensitivity models.

After all feature engineering, the final analytic data frame contains **1,449 columns** and 22,431 rows.

Logistic Regression Results — Refined Model (Top 10 % Outcome)

Using the top-10 % binary outcome and categorical treatment via `C()` notation in statsmodels, a refined logistic regression was estimated. Model fit: **Pseudo $R^2 = 0.131$**; N = 22,431.

Table 3. Odds Ratios — Full Sample, Refined Model (Top 10 % Outcome)

Predictor	OR	95 % CI	p-value
Age (per year)	1.022	1.019–1.025	<0.001
Female sex (vs. Male)	1.186	1.080–1.303	<0.001
Race 2 (vs. Race 1)	0.817	0.711–0.939	0.004
Race 4 (vs. Race 1)	0.579	0.443–0.757	<0.001
Region 1 (vs. ref)	0.152	0.102–0.226	<0.001
Region 2	0.147	0.100–0.218	<0.001
Region 3	0.122	0.083–0.180	<0.001
Region 4	0.120	0.081–0.178	<0.001
Poverty Cat. 4 (vs. ref)	0.776	0.668–0.901	0.001
Uninsured (INSCOV22=3)	0.194	0.126–0.298	<0.001
Multimorbidity	**2.930**	2.621–3.275	**<0.001**

Table 4. Multimorbidity OR by Gender — Top 10 % Outcome

Sub-Group	OR (Multimorbidity)	95 % CI
Full Sample	2.930	2.621–3.275
Male	2.959	2.487–3.520
Female	2.914	2.521–3.369

Conclusion

Using 2022 MEPS data, this analysis demonstrates that **multimorbidity is the dominant predictor of high healthcare expenditure risk**. Individuals with two or more chronic conditions have approximately **3× the odds** of high annual expenditure compared to those without, after adjusting for age, sex, race, region, poverty, and insurance status. This effect is consistent across males and females. Pronounced regional variation and the suppressive effect of being uninsured on recorded expenditure are also notable findings.

References

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